



TJ-MU-10B

Pump and mixing unit for underfloor heating
manifolds · 210 / 235 mm · 20–43 °C



termojet.com.ua

WARNING! Read this manual before putting the product into service. For safety reasons, only persons familiar with its contents may install or service the product.

Application

The Termojet TJ-MU-10B pump and mixing unit is designed for mounting on underfloor and radiator heating manifolds with a port-to-port spacing of **210 or 235 mm**. The unit forms a separate low-temperature circuit: a thermostatic valve blends cooled return water into the hot flow and holds the water entering the floor loops at a constant temperature.

The unit is used on manifolds requiring a flow rate of no more than **2.5 m³/h**. Control is entirely passive — no power supply, no external sensors, no capillary tubes.

Scope of supply

- thermostatic mixing valve 20–43 °C, Kvs 2.5 m³/h, with a Vernet wax thermostatic element;
- union nuts for a G1½" pump with a 130 mm port-to-port length;
- thermometer with a 0–80 °C scale;
- return tee with a built-in **non-return valve**;
- 1" nipples for connection to the flow and return bars of the manifold.



Upper part: thermometer and the connection to the flow bar



Lower part: thermostatic valve and the tee with the non-return valve

How it works

The boiler or heat pump delivers water at a temperature higher than the screed can take. The pump of the unit circulates the water through the floor loops continuously, while the thermostatic valve compares the temperature of the mixed water at the outlet against the value set on the handle and changes the ratio of hot flow to cooled return accordingly. The wax element expands and contracts inside the valve body and moves the valve disc mechanically — the unit needs no power supply and no external control.

The circulating pump is not supplied with the unit — it is ordered separately, see p. 3.

Technical data

Temperature setting range	20–43 °C
Maximum working temperature	95 °C (110 °C short-term)
Mixing accuracy at the outlet	± 2 °C
Recommended working pressure	3 bar
Maximum static pressure	10 bar
Maximum working pressure	5 bar
Maximum pressure difference between inlets	3 bar
Flow coefficient of the valve, Kvs	2.5 m³/h
Medium	water, water with up to 30 % glycol
Connection to the manifold	1" male thread
Pump connection	G1½", 130 mm port-to-port
Port-to-port spacing of the unit	210 or 235 mm
Valve body	CW617N brass
Handle	nylon
Seals	EPDM
Thermostatic element	wax, Vernet (France)

Selecting the circulating pump

The pump is not supplied with the unit. The unit takes pumps of size 25-40/130, 25-60/130 or 25-80/130 — with a G1½" connection and a port-to-port length of 130 mm.

Recommended models:

Part no.	Pump
30254013	Termojet AUTO APE 25/40/130
30256013	Termojet AUTO APE 25/60/130
30258013	Termojet AUTO APE 25/80/130

Dimensions

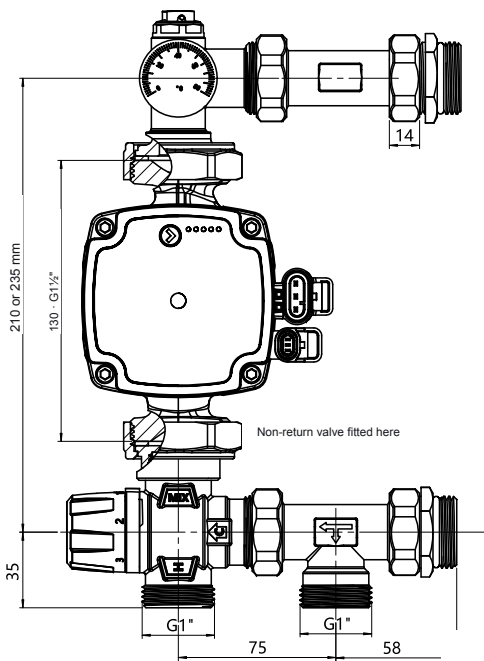


Fig. 1. The unit assembled with a pump, mm

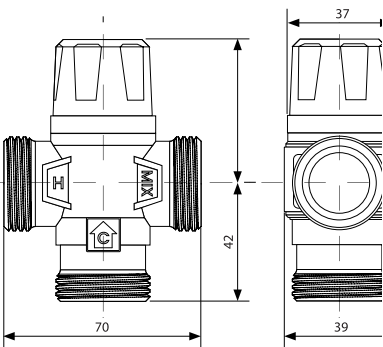


Fig. 2. Thermostatic valve, mm

The pump in Fig. 1 is shown for reference only — it is not supplied with the unit.

Application example

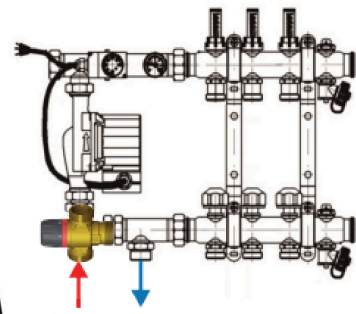
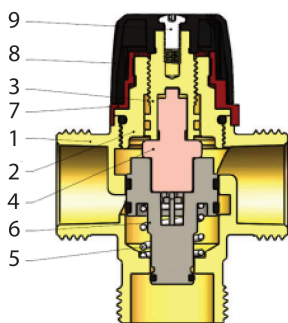


Fig. 3. The unit holds a constant flow temperature in an underfloor heating system

Valve construction



- 1. Valve body: CW617N
- 2. Valve disc: CW617N
- 3. Adjustable valve: CW617N
- 4. Wax element: Vernet
- 5. Spring: SUS304
- 6. Water distributor: PSU
- 7. Position indicator: nylon
- 8. Handle: nylon
- 9. Handle screw

Temperature setting

Scale position	Temperature, °C
below «1» (minimum)	18
1	20
2	25
3	30
4	35
5	43
6 (maximum)	46

The outlet temperature is set with the handle according to the table. To change the setting, release the handle by pulling it gently upwards. After the change, lock the handle again — this keeps the setting from being knocked off accidentally.

The dots between the figures on the scale are intermediate positions. The actual flow temperature also depends on the inlet water temperature and on the flow rate in the circuit, so check it against the thermometer of the unit after commissioning.

Installation notes

- Installation is carried out by qualified trades.
- Before fitting, clean the pipework of scale and rust; a strainer at the system inlet is required for trouble-free operation.
- The direction of flow must match the arrows on the body.
- Fit the pump so that its shaft is horizontal.
- After installation, test the system for leaks.
- Check the operation of the valve at least once every 12 months.

Using pipe wrenches during installation is not permitted.

The product falls under Directive 2014/68/EU on pressure equipment and, in accordance with Art. 4.3 (sound engineering practice), does not carry the CE marking.

Product data

Product	Termojet TJ-MU-10B pump and mixing unit
Manufacturer	Termojet
Distributor in Ukraine	Sofiivka-Montazh LLC, EDRPOU 37074476 EDRPOU — Ukrainian company registry code
Address	3 Kyivska St., Sofiivska Borshchahivka, Bucha district, Kyiv region, 08131, Ukraine
Contacts	+380 44 407 24 10 · termojet@sofiivka.kiev.ua termojet.com.ua

Storage and transport

Store the product in the manufacturer's packaging, in heated or unheated warehouses — no closer than one metre to any heating appliance — or under a canopy, in conditions that rule out mechanical damage.

Permissible storage and transport temperature is –20 to +50 °C. The product may be carried by any means of transport in accordance with the carriage rules in force for that mode of transport. During carriage, protect the product from impact, mechanical load and surface scratching.

Disposal

The product contains no substances hazardous to human life and health or to the environment. At the end of its service life, dispose of the product as household waste in accordance with local waste-management rules. Metal parts are suitable for remelting.

Acceptance and testing

The products covered by these instructions have been manufactured, tested and accepted in accordance with the manufacturer's current technical documentation.

Warranty

The manufacturer warrants that the product complies with safety requirements, provided that the user observes the rules set out in these instructions.

The service life of the product, provided that these instructions are observed and the required servicing is carried out, is **10 years** from the date the product is handed over to the user.

The warranty period is **24 months** from the date of sale, but no longer than the service life of the product.

The warranty covers all defects arising through the fault of the manufacturer and **does not cover** defects caused by:

- failure to observe the storage, installation, testing, operation or servicing conditions described here;
- improper transport or loading and unloading;
- exposure to substances aggressive to the materials of the product;
- fire, natural disaster or force majeure;
- acts of the user or third-party interference with the design of the product.

Products that fail within the warranty period because of a manufacturing defect are repaired or replaced free of charge. The cost of removing and transporting the faulty product is not reimbursed to the buyer. If a claim proves to be unfounded, the buyer pays for the diagnosis and examination.

To process a claim the following are required: a free-form statement giving contact details, the installation address and a description of the defect; proof of purchase; photographs of the faulty product, including photographs taken at the place of installation; a copy of the completed warranty card.

Claims are accepted at: 3 Kyivska St., Sofiivska Borshchahivka, Bucha district, Kyiv region, 08131, Ukraine. Phone +380 44 407 24 10, e-mail termojet@sofievka.kiev.ua.

The manufacturer reserves the right to introduce changes to the design of the «Termojet TJ-MU-10B pump and mixing unit» product that do not impair its quality.

Warranty card

to delivery note No. _____ dated _____ 20____

Product: Termojet TJ-MU-10B pump and mixing unit

No.	Part number	Quantity	Notes
1			
2			
3			
4			

Warranty period – 24 months from the date of sale.

I have read and accept the warranty terms and the installation and operating instructions:

Buyer (signature)

Seller (signature)

Date of sale

Stamp or seal
of the selling organisation